

Sami Aljabery

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Education

M.S. in Electrical and Computer Engineering

Oregon State University | Corvallis, OR

Expected: Jun. 2026

GPA: 3.67

- Concentration in Computer Architecture

B.S. in Electrical and Computer Engineering

Oregon State University | Corvallis, OR

Dec. 2024

GPA: 3.51

- Minor in Computer Science

Relevant Coursework

- CMOS Integrated Circuits, Contemporary Energy Applications, Analog CMOS Integrated Circuits
- VLSI System Design, Energy Efficient VLSI Design, Microcontroller System Design
- High Performance Computer Architecture, Modern Memory Systems, Introduction to Parallel Programming
- Computer Architecture, Operating Systems II, Advanced Deep-Learning & High Performance Computing

Experience

Graduate Teaching Assistant

Oregon State University | Corvallis, OR

Jan. 2025 – Present

- Guided 100+ students in ECE Capstone, Computer Networks, and Computer Organization & Assembly through lectures, labs, and office hours.
- Facilitated students in applying tools such as oscilloscopes, frequency generators, power supplies, Wireshark, and debugging AVR assembly to reinforce concepts in circuit design, computer networks, and microcontroller programming.
- Collaborated with instructors and teaching assistants to refine course material, improving student learning outcomes.

Digital Design Engineering Intern

onsemi | Corvallis, OR

Jun. 2025 – Sept. 2025

- Implemented python-based scripts to automate verilog validation tests of non-deterministic random bit generator in the AR0827 image sensor, assessing entropy quality and oscillator reliability for hardware security.
- Devised and assembled a PCB to synchronize two image sensors via raspberry pi and logic level shifting for High-Speed Shutter validation in DevWareX.
- Migrated existing HTML server to confluence and developed a reusable quiz generator, streamlining creation of training content for new application engineers.

EECS Undergraduate Learning Assistant

Oregon State University | Corvallis, OR

Mar. 2022 – Dec. 2024

- Delivered weekly recitations for 60+ students in Electrical Fundamentals and Digital Logic Design.
- Demonstrated hands-on use of oscilloscopes, power supplies, and waveform generators to reinforce lab concepts.
- Taught students FPGA design and simulation with Intel Quartus Prime, developing essential digital logic skills.

Physics Undergraduate Learning Assistant

Oregon State University | Corvallis, OR

Mar. 2022 – Aug. 2022

- Assisted in recitations for 30+ students in General Physics with Calculus, reinforcing core concepts such as kinematics.
- Cooperated with three instructors to refine course content, improving clarity and accessibility to better meet student learning needs.

Projects

Multi-Sensor Monitoring System

Nov. 2024 – Dec. 2024

- Developed a real-time monitoring system in C on a custom Atmega 32U4 board, writing custom drivers to interface

with an I2C thermal sensor and onboard LCD.

- Established an interrupt-driven timing system by configuring two hardware timers, one set to CTC mode to generate a real-time clock and one in input capture to measure ultrasonic sensor echo pulses for distance calculations.
- Integrated multiple hardware peripherals, including an on-board and external temperature sensor, an external ultrasonic sensor, and 4 push buttons for with software debouncing for user-controlled clock adjustment and LCD backlight brightness.

FPGA-Based Gameboy

Sept. 2023 – Jun. 2024

- Led a team of 4 to develop a portable, handheld Gameboy on a DE10-nano using the open-source MiSTer project, managing project integration and software boot-up.
- Engineered and integrated external hardware to optimize portable display and validated gameboy opcodes using Modelsim testbenches written in Verilog.
- Developed a compact, ergonomic enclosure using Autodesk Fusion 360, achieving a 10% size reduction from the initial assembly.

Pipelined Microprocessor

May 2024 – Jun. 2024

- Created a 16-bit microprocessor in Verilog, modeling the instruction fetch (IF) stages of a 5-stage pipeline to manage instruction memory and the program counter.
- Implemented a 16-register bank and instruction-decoding logic within the execute stage to parse 16-bit instructions and route operands to specialized processing units.
- Validated custom ALU and SRU modules, creating custom Verilog testbenches in Xilinx Vivado to verify correct execution of 8 operations, including ADD, NAND, XOR, and ROTL.

4-bit Arithmetic Logic Unit

Feb. 2024 – Mar. 2024

- Designed, simulated, and validated a 4-bit transistor-level ALU and accumulator through Cadence Virtuoso.
- Established arithmetic and logic operations using a 32:4 multiplexer and custom PLA for instruction decoding.
- Analyzed sequential performance by integrating the adder into an accumulator with D flip-flops, identifying the critical carry path and determining the most energy-efficient maximum clock frequency (1 GHz at 1.9V) through transient analysis.

Hobbyist Oscilloscope | Embedded Systems Developer

Apr. 2023 – Jun. 2023

- Partnered with 2 peers to develop a portable oscilloscope utilizing a high sampling microcontroller.
- Programmed data logging to SD cards via C++, enabling waveform analysis in .csv format.
- Built dual rotary encoder interface for intuitive voltage and time control, enhancing the user interface.

Leadership

Embedded Systems Design Club | Club President

Aug. 2024 – Present

- Founded the Embedded Systems Design Club (ESDC), hosting more than 20 meetings, workshops and projects that build practical embedded systems skills including hardware knowledge, communication protocols, and debugging.

Awards & Recognition

Dean's List - Oregon State University

Jun. 2022 – Present

Engineering Excellence Award - Oregon State University

Jun. 2024

Skills

Languages: C++, C, Verilog, SystemVerilog, Python, AVR Assembly, LaTeX, OpenMP, OpenCL, CUDA, HTML

Tools: Cadence Virtuoso, Autodesk Fusion 360, LTSpice, KiCAD, Quartus Prime, Xilinx Vivado, MATLAB, Linux

Technical: Oscilloscope, Microcontrollers, FPGAs, Operating Systems, PCB Design, Soldering, Git

Protocols: SPI, I2C, UART, MPI